

SANAM RAI

Backend Developer · MERN Stack · Kathmandu, Nepal

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PERSONAL PROFILE

I am BIT graduate from Bhaktapur Multiple Campus affiliated with Tribhuvan University. I am very passionate about building backend systems as well as design a UI frontend system. I have built and deployed some projects from scratch and shipped to production. I am also former QA intern with hands on testing, debugging and documentation experience. I also have completed Harvard CS50x and CS50P and hold their certification. I am eager to contribute to the real world enterprise projects and grow under experienced mentors.

Live Portfolio: sanam-rai.com.np

TECHNICAL SKILLS

Languages: Python, JavaScript (ES6+), HTML5, CSS3

Backend: Node.js, Express.js, Flask, REST API Design, JWT Authentication, Socket.IO, bcrypt, Rate Limiting, Morgan, Response Time Middleware

Frontend: React.js, React Context API, Custom Hooks, React Router v6, Axios Interceptors, Tailwind CSS, Responsive UI Design, Skeleton Loading

Databases: MongoDB (Atlas), Mongoose (populate, aggregation), MySQL

Testing & QA: Selenium (Python), Playwright (Python), Pytest, Manual Testing, Functional & Regression Testing, Bug Reporting

Tools: Git, GitHub, Postman, VS Code, Linux

Deployment: Render (Web Services), Vercel (Frontend), MongoDB Atlas (Cloud DB)

PROJECTS

Krishi Bazar Agricultural E-Commerce Marketplace

Node.js · Express.js · React.js · MongoDB · JWT · Tailwind CSS · Context API · Render · Vercel

Live: krishi-bazar-alpha.vercel.app [GitHub](#)

Built and deployed a complete full-stack agricultural marketplace where Nepali farmers sell produce directly to buyers covering product listing, search, category filtering, sorting, and pagination.

- Implemented dual checkout flows: cart-based checkout (fetch from MongoDB, populate product details) and Buy Now (direct single-product checkout) both handled by a single unified order controller.
- Designed a layered Context architecture AuthContext (JWT + user state), CartContext (cart CRUD synced with backend), NotificationContext (global in-app alerts) with custom hooks for clean consumption.
- Built a notification system from scratch using React Context and setTimeout independently arrived at the same architecture used by react-hot-toast and react-toastify internally.
- Implemented skeleton loading on all data-heavy pages using CSS shimmer animation separate skeleton components mirror real card shapes for a polished perceived performance.
- Configured Axios interceptor to read JWT from localStorage and attach it to every protected request automatically eliminating manual header management across the entire frontend.
- Secured cart and checkout routes with JWT middleware (backend) and PrivateRoute component (frontend) unauthenticated requests return 401, UI redirects to login.
- Used Mongoose populate to resolve product references inside cart items returning full product details (name, price, image) alongside cart quantities in a single query.
- Deployed backend on Render, frontend on Vercel, and database on MongoDB Atlas configured CORS for cross-domain communication and tested the full flow on production.

Portfolio System Feature Toggle Architecture

Node.js · Express.js · MongoDB · JWT

Live: sanam-rai.com.np [GitHub](#)

- Built a backend-driven portfolio API where system features JWT auth, in-memory caching, rate limiting, pagination, and request logging can each be toggled on or off independently at runtime without restarting the server.
- Implemented JWT-based authentication middleware with toggle control, allowing production-like API protection to be demonstrated and disabled for testing without code changes.
- Designed an in-memory caching layer that reduces database round-trips on repeated queries cache invalidation handled on toggle disable.
- Developed custom pagination logic with query-based page navigation and total count metadata for scalable data retrieval.
- Structured entirely with clean modular architecture controllers, middleware, routes, and config fully separated for maintainability.

YakTalk Secure Real-Time Chat Application

Node.js · Express · Socket.IO · React · MongoDB · JWT · Tailwind CSS · Render

Live: yaktalk-chatapp-1.onrender.com [GitHub](#)

- Developed and deployed a full-stack real-time messaging application with private chat, online presence indicators, and JWT-based session authentication.
- Secured WebSocket connections by verifying JWT tokens server-side during Socket.IO handshake preventing unauthorised real-time access before the connection is established.
- Resolved CORS and cross-domain cookie challenges for credentials-based communication between separately deployed frontend and backend services on Render.
- Implemented bcrypt password hashing and secure session management across cross-origin deployments.

Phone Book Application

Python · Pytest · Regex · JSON [GitHub](#)

- Built a CLI-based contact management system with full CRUD operations — add, view, update, and delete contacts.
- Used regular expressions for input validation and JSON for persistent local data storage.
- Applied QA principles from internship to personal project wrote Pytest unit tests to improve reliability.

PROFESSIONAL EXPERIENCE

Quality Assurance Intern

Dec 2025 – Mar 2026

Danfe Solution Pvt. Ltd. · Kathmandu, Nepal

- Conducted manual functional testing across multiple internal systems, identifying and documenting defects with clear reproduction steps.
- Performed regression and validation testing to ensure application stability after feature updates and patches.
- Collaborated closely with developers to debug issues, triage defects, and improve overall product quality.
- Created technical documentation to streamline QA workflows and standardise internal processes.
- Assisted in client onboarding and product training sessions, improving user adoption and product understanding.

EDUCATION

Bachelor in Information Technology (BIT)

Completed (Certificate Provisionally Waiting) · Aggregate CGPA: [3.1]

Tribhuvan University - Bhaktapur Multiple Campus · Nepal

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Software Engineering, Operating Systems, Computer Networks

+2 / Higher Secondary Education

GPA: [3.1]

Xavier International school · Nepal

SLC / SEE (Secondary Education Examination)

GPA: [3.60]

Boston International school · Nepal

CERTIFICATIONS

- [CS50x: Introduction to Computer Science Harvard University \(edX\)](#)
- [CS50P: Introduction to Programming with Python Harvard University \(edX\)](#)